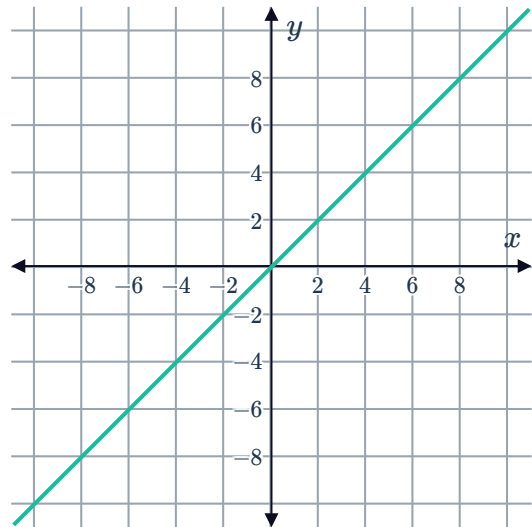


Gradient function graphs



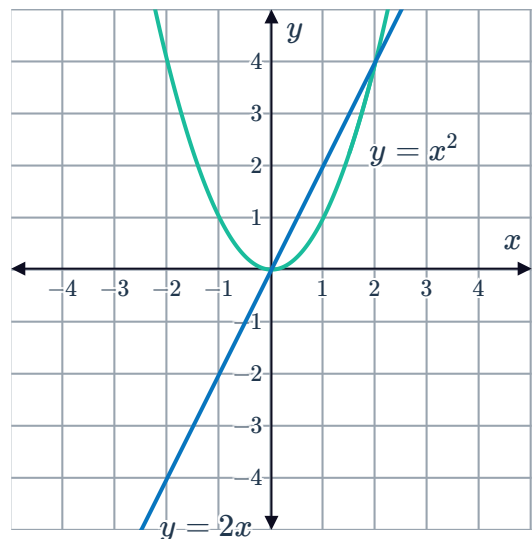
1 Consider the graph of $y = x$:

- a Find the gradient of the line at $x = 4$.
- b Find the gradient at any value of x .
- c What can be said about the gradient of a linear function?



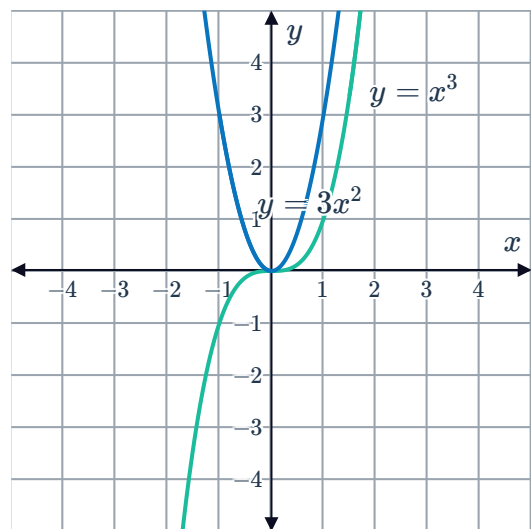
2 Consider the graph of $y = x^2$ and its gradient function $y = 2x$:

- a What can be said of the sign of the gradient function when $y = x^2$ is increasing or decreasing?
- b For $x > 0$, is the gradient of the tangent positive or negative?
- c For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?
- d For $x < 0$, is the gradient of the tangent positive or negative?
- e For $x < 0$, as the value of x increases how does the gradient of the tangent line change?
- f For $y = x^2$, the gradient of the tangent line changes at a constant rate. What type of function is the derivative of $y = x^2$?



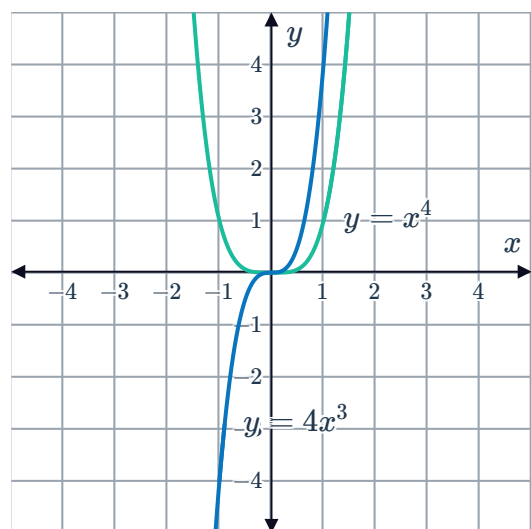
3 Consider the graph of $y = x^3$ and its gradient function $y = 3x^2$.

- For $x > 0$, is the gradient of the tangent positive or negative?
- For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?
- For $x < 0$, is the gradient of the tangent positive or negative?
- For $x < 0$, as the value of x increases how does the gradient of the tangent line change?
- For $y = x^3$, the gradient of the tangent line first decreases at a decreasing rate, then increases at an increasing rate. What type of function is the derivative of $y = x^3$?



4 Consider the graph of $y = x^4$ and its gradient function $y = 4x^3$.

- For $x > 0$, is the gradient of the tangent positive or negative?
- For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?
- For $x < 0$, is the gradient of the tangent positive or negative?
- For $x < 0$, as the value of x increases how does the gradient of the tangent line change?
- For $y = x^4$, the gradient of the tangent line is increasing, first at a decreasing rate and then at an increasing rate. What type of function is the derivative of $y = x^4$?



5 Consider the functions $f(x) = x^5$ and $g(x) = x^4$.

- Sketch the graph of $f(x)$ and its derivative.
- Sketch the graph of $g(x)$ and its derivative.
- If the degree of a function is even, will the degree of its derivative function be odd or even?
- If the degree of a function is odd, will the degree of its derivative function be odd or even?



Algebraic differentiation

- 6
- a For each of the following functions, state the degree of the gradient function:
- i $y = x^2$ ii $y = x^3$ iii $y = x^4$
- b Hence, state the degree of the derivative of a polynomial function of degree n .
- 7 Find the derivative of the following functions with respect to x :
- a $y = x^7$ b $y = x^9$ c $y = x^5$ d $y = -6$

Applications

- 8 Find the gradient of $f(x) = x^4$ at $x = 2$. Denote this gradient by $f'(2)$.
- 9 Consider the function $f(x) = x^9$.
- a Find the gradient of the tangent to the function at $x = 1$.
- b Find the equation of the tangent to the function at $x = 1$.
- 10 David draws the graphs of x^2 , x^3 , x^4 and x^5 and draws the tangents to each one at the point where $x = 1$.
- a David then notices where each of the tangents cut the y -axis and records this in the table below:

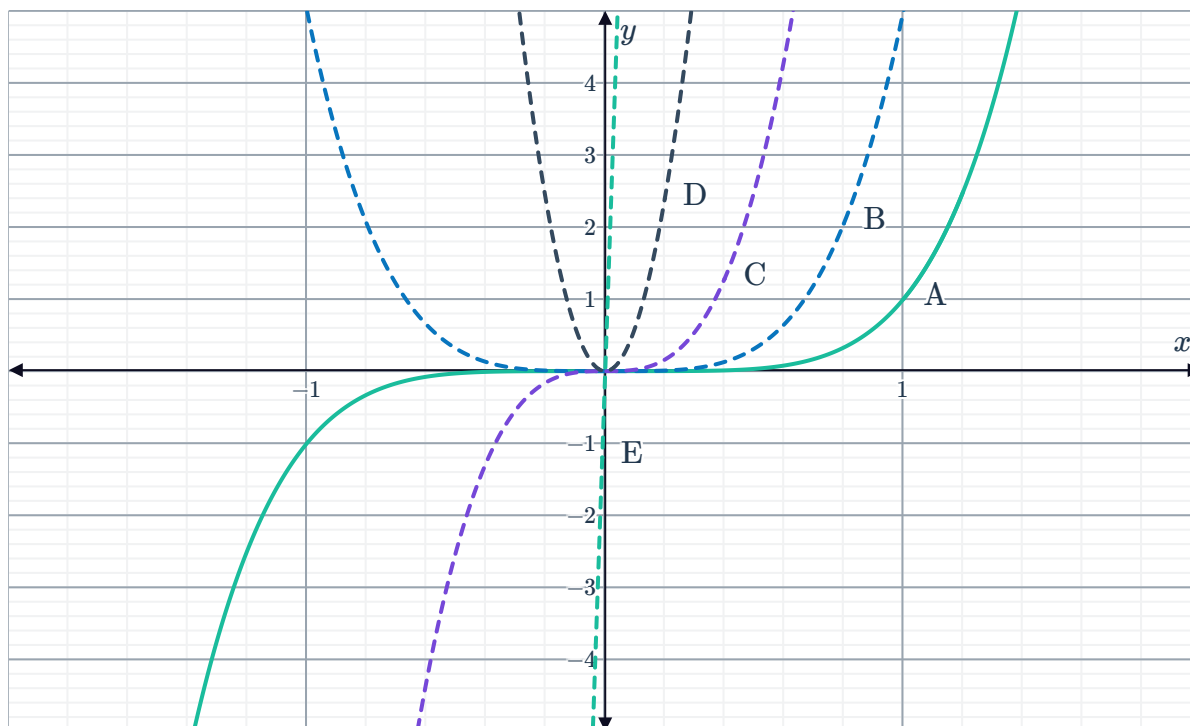
Graph	y -intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	
$y = x^3$	$(0, -2)$	
$y = x^4$	$(0, -3)$	
$y = x^5$	$(0, -4)$	

Complete the table by calculating the gradient of each of the tangents at $x = 1$.

- b Following the pattern in the table, what would be the gradient of the tangent to the graph of $y = x^n$ at the point where $x = 1$?



- 11 The graph of $y = x^5$ is shown below labelled  A. Fiona then graphs the derivative of the function, labelling it as B. She then finds the derivative of graph B to get graph C, then differentiates again to get graph D and differentiates again to get graph E.



State whether the following statements are true about this sequence of derivatives:

- a The derivative of a function is always positive when the function is negative, and negative when the function is positive.
- b Each graph is a function of the form ax^n .
- c For any value of x , the value of the derivative will always be greater than the value of the function.
- d The degree of the derivative is always different to the degree of the function.